

Ashwin Kumar S

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ABOUT ME

I am a 3rd year student who is deeply passionate about software development. My expertise spans various projects encompassing Python, Flutter, C++, Java and MERN stack. My journey in app and web development has been primarily self-taught through hands-on experiences and project endeavors. I am also familiar with machine learning and have created solutions for modern problems. Driven by my creativity and thoughts of optimizations in code using DSA, I would love to create innovative and optimal software solutions.

PROJECTS

Health Monitoring System | Wear OS | Kotlin | Android Studio | Firebase

- Developed a Wear OS application using Kotlin within Android Studio, aimed at monitoring user activity and promoting a healthy lifestyle with health tips of 90 percent accuracy and usefulness.
- Implemented features to send users notifications for dietary reminders 4 times a day and to track their medical records seamlessly.
- Utilized Firebase as a backend service for secure storage and retrieval of user data, ensuring privacy and accessibility across devices.

Digitization of data for Student Extracurricular activities | MySQL | Python Flask | Rest APIs

- A website for the college students and teachers where extracurricular records and proofs of students are managed and easily accessed.
- This digitization allows 75% of work to be managed digitally modernizing the data management system.
- Query optimization with proper DDL and DML queries improves performance by 40 percentage.

ICCCSP 2025 | Next.js | SASS(SCSS) | Tailwind CSS | CPanel | [ICCCSP](#)

- Designed and developed the official website for the 9th International Conference on Computer, Communication, and Signal Processing in collaboration with SSN, Elsevier, and Procedia for paper publications.
- Implemented APIs and processing algorithms to enhance website efficiency and optimize performance.
- Utilized Next.js and Sass for responsive design and dynamic content, deployed seamlessly on cPanel for robust hosting.

Hotel Reservation System | Data Structures | Python | Tkinter

- The project aims to allocate tables in hotels and restaurants effectively reducing the crowd.
- The system is implemented with 6-hardcoded data structures like Hash Table, Queue, Linked List allowing efficient data storage and retrieval.

Thought Controlled Car Steering using EEG-Brain Waves | Raspberry pi | Flask | Bi-LSTM

- By using the EEG signals and proper signal processing techniques, a deep learning model which uses Bi-LSTM layers is trained such that 95.6% accuracy is achieved.
- A research paper describing the methods and model is also written marking its completion.

Mentor-Student management | Python Flask | Design Patterns | Oracle SQL

- Developed a Mentor-Student management system using Python Flask framework, implementing efficient communication and collaboration between mentors and students.
- Employed design patterns to ensure scalability, maintainability, and extensibility of the system's architecture which improves work efficiency by 60%.
- Integrated Oracle SQL database for robust data management and backend management using flask to integrate database and user interface.

Education

B.TECH- INFORMATION TECHNOLOGY

Sri Sivasubramaniya Nadar College of Engineering

CGPA – 8.97 | Department Rank - 3 | Graduating 2026 | Chennai, Tamil Nadu

Internship & Experience

IITM Research Park - HTIC (Healthcare and Technology Innovation Center)

Software Developer Intern

- Reworked on their software GUI application with 3000+ lines of code to add desired functionalities and optimizations in C++ using the QT framework and VTK (Visual Tool Kit) Library.
- Made a software to software API in C++ that takes in python scripts for the software and run them in background without disturbing the application.

App Development Sub-Head at ProCoDe-(Intra Department Coding Club).

Primary developer for website of INVENTE 2024, the annual technical fest of SSN College.

Student Coordinator for Tech-Grad School(1 week Technical event in department).

Skills

Programming Languages

Python, C, C++, Dart (Flutter), Java, JavaScript.

Development

HTML5/CSS3, Python Flask & Django, React, QT Creator(C++), Ruby on Rails, Node.js, Express.js, Next.js, Tailwind CSS, JIRA, Kotlin, Arduino and ESP32, Regression and CNN Models.

Databases and others

MYSQL, Oracle SQLPlus, MongoDB, Firebase, Supabase, Data structures and Algorithms, Unit Testing, Design Patterns and UML Modeling

Certifications

Ethical Hacking - NPTEL, Flutter Essential Training - LinkedIn, Supervised Machine Learning - Stanford Online(Coursera), Programming and Problem Solving in C - NPTEL, SQL Databases - Hackerrank

Awards

- Received Merit Scholarship for 2nd Rank in Department in 2023.
- Received a Silver Medal for 2nd Highest GPA in 1st Semester on College Day.
- Received Merit Scholarship for 3rd Rank in Department in 2024.
- Solved 500+ problems in leetcode and have a rating on 1700 in contests.